

Storage & Memory performance

1. Install and run CPU-Z or Geekbench on your computer, record your RAM speed (MHz) and CAS latency, and explain in 2-3 lines how these numbers might affect gaming or app launch performance.

-> RAM Speed and CAS Latency

Run **CPU-Z** → **Memory** tab and record the values shown:

- **RAM Speed:** _____ MHz (CPU-Z shows **DRAM Frequency**, which is about half the effective DDR speed)
- **CAS Latency (CL):** _____ cycles

Example: If CPU-Z shows **1600 MHz** DRAM Frequency and **CL16**, the effective memory speed is approximately **DDR4-3200**.

2–3 Line Explanation

Higher RAM speed can improve memory bandwidth and may help gaming and applications respond more quickly, particularly in memory-sensitive workloads. Lower CAS latency generally means less delay before the RAM begins responding, although real-world performance depends on both speed and latency along with the CPU and other system components.

2. Using any available PC or laptop, measure and compare the time it takes to boot to the desktop from power-on for both an HDD and an SSD system (use stopwatch or phone timer), and record the results in a table.

Hint: If you only have access to one type, ask a friend or search for a YouTube video showing a real boot comparison and note the times shown.

-> HDD vs SSD Boot-Time Comparison

Use a stopwatch or phone timer and measure from **pressing the power button until the desktop is fully loaded and usable**. Ideally, perform **3 tests for each drive** and calculate the average.

Test	HDD Boot Time	SSD Boot Time
Test 1	____ seconds	____ seconds
Test 2	____ seconds	____ seconds
Test 3	____ seconds	____ seconds
Average	____ seconds	____ seconds

How to perform the test

1. Shut down both systems completely.
2. Start the stopwatch immediately when you press the power button.
3. Stop it when the desktop has finished loading and is ready to use.
4. Repeat three times for more reliable results.
5. If you only have one type of drive, use a friend's PC/laptop or a real-world boot-comparison video and clearly label the source of the second result.

Conclusion template:

The SSD system took approximately ____ **seconds** to boot, while the HDD system took approximately ____ **seconds**. The SSD completed the boot process faster because SSDs have much lower access latency and faster random-read performance than mechanical HDDs.

3. Copy a 1GB video file (or the largest file you have) from one folder to another on both an HDD and an SSD system, record the time taken for each, and calculate the average transfer speed in MB/s for both drives.

-> HDD vs SSD File Transfer Test

Use the **same video file** and copy it between folders on each system. Record the file size and the time required.

Formula:

Transfer speed (MB/s) = File size (MB) ÷ Transfer time (seconds)

Drive	File Size	Transfer Time	Average Speed
HDD	1,024 MB	_____ seconds	_____ MB/s
SSD	1,024 MB	_____ seconds	_____ MB/s

Example Calculation

If the 1 GB file takes **20 seconds** on the HDD:

$$1,024 \div 20 = 51.2 \text{ MB/s}$$

If it takes **5 seconds** on the SSD:

$$1,024 \div 5 = 204.8 \text{ MB/s}$$

Procedure

1. Use the same 1 GB video file for both tests.
2. Copy it from one folder to another on the **HDD** and record the time.
3. Repeat the test on the **SSD**.
4. Ideally, perform **3 tests per drive** and calculate the average transfer time/speed.
5. Make sure the source and destination are on the same type of drive when comparing the drive's performance.

Conclusion: An SSD will generally provide higher file-transfer performance than an HDD, although actual speeds depend on the specific drive, interface, file size, and system workload.

4. Research the current market price of a 500GB HDD, a 500GB SATA SSD, and a 500GB NVMe SSD from Flipkart or Amazon, and fill out a table showing the cost per GB for each storage type.

-> Here is a current India price comparison based on listings I found. Prices can change with sales and sellers.

Storage Type	Example	Price	Cost per GB
500GB HDD	Seagate 500GB SATA HDD	₹2,449	₹4.90/GB
500GB SATA SSD	Crucial BX500 500GB	₹9,999	₹20.00/GB

Storage Type	Example	Price	Cost per GB
500GB NVMe SSD	Crucial P3 500GB NVMe	₹3,500	₹7.00/GB

The 500GB HDD listing is currently shown at ₹2,449 on Flipkart. ([Flipkart](#)) The Crucial P3 500GB NVMe listing on Amazon is ₹3,500. ([Amazon India](#)) Flipkart's current 500GB SSD listings also show several SATA/NVMe options, with prices varying considerably by model and seller. ([flipkart.com](#))

Formula: Cost per GB = Price ÷ 500.

5. Based on your benchmarks and cost analysis, write real-world storage recommendations for three scenarios: (1) a budget gaming PC, (2) a college student's laptop for coding and Netflix, (3) a high-end video editing workstation. Justify your choice for each scenario in 2-3 lines.

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Scenario	Recommended Storage	Justification
1. Budget Gaming PC	500GB SATA SSD	A SATA SSD offers much faster boot and game-loading times than an HDD while keeping the cost relatively low. It is a good balance of performance and price for a budget gaming PC.
2. College Student – Coding & Netflix	500GB NVMe SSD	An NVMe SSD provides fast application launches, coding-project access, and system boot times. 500GB is enough for development tools, documents, and a reasonable amount of media without needing a large HDD.
3. High-End Video Editing Workstation	1TB–2TB NVMe SSD + large HDD for storage	NVMe provides very high read/write speeds for video editing, project files, and media-cache operations. A large HDD can be added for storing completed projects and backups at a lower cost per GB.

Conclusion: For most users, an SSD is preferable to an HDD for the operating system and applications because its much lower access latency improves responsiveness. For professional video editing, combining **NVMe performance with high-capacity HDD storage** provides both speed and economical capacity.

